Theoretical Physics 2 - Blatt 04

Tobias Laser

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10. Measurement Probability Given the three-dimensional wavefunction

$$\psi(x,y,z) = N \exp\left(-\left(\frac{|x|}{2a} + \frac{|y|}{2b} + \frac{z}{2c}\right)\right)$$

where a, b, c are positive real numbers.

(a) Compute the normalization constant N.

Calculation

$|N|^2 \int dxdydz \exp\left(-\left(\frac{|x|}{2a} + \frac{|y|}{2b} + \frac{z}{2c}\right)\right) = 1$

Calculation

$\int\limits_0^{+\infty} dx \exp\left(-\frac{x}{a}\right)$

$|N|^2\, 2^3 abc = 1 \Leftrightarrow N = \frac{1}{\sqrt{2^3 abc}}$

(b) What is the probability that a measurement of X yields a result between 0 and a?

Calculation

$\int\limits_0^a dx \frac{1}{2^3 abc} \exp\left(\frac{-x}{a}\right) \int dydx \exp\left(-\left(\frac{|y|}{b} + \frac{|z|}{c}\right)\right) = \int\limits_0^a dx \frac{\exp\left(-\frac{x}{a}\right)}{2a} = -\frac{a \exp\left(-\frac{x}{a}\right)}{2a} \Big|_0^a = \frac{1}{2}(1 - \frac{1}{e})$

(c) What is the probability that a simultaneous measurement of Y and Z yields values between $-b$ and b, and between $-c$ and c, respectively?

Calculation

$\int\limits_{-\infty}^{+\infty} dx \int\limits_{-\infty}^{+\infty} dy \int\limits_{-\infty}^{+\infty} dz \frac{1}{2^3 abc} \exp\left(-\left(\frac{|x|}{a} + \frac{|y|}{b} + \frac{|z|}{c}\right)\right) =$

$\frac{4}{4bc} \int\limits_0^b dy \exp\left(-\frac{y}{b}\right) \int\limits_0^c dz \exp\left(-\frac{z}{c}\right) = \frac{1}{bc}\left(-b \exp\left(-\frac{y}{b}\right)\right) \Big|_0^b \left(-c \exp\left(-\frac{z}{c}\right)\right) \Big|_0^c = (1 - \frac{1}{e})^2$

(d) Compute the probability that a measurement of the momentum yields a result within the volume element $dp_x dp_y dp_z$, centered at the point $p_x = p_y = 0, p_z = \frac{\hbar}{c}$.

Calculation

Probability amplitude in momentum space:

$$\phi(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int dx\, dy\, dz\; \psi(x,y,z)\, e^{-i\frac{p_x x + p_y y + p_z z}{\hbar}}$$

For $p_x = p_y = 0$, $p_z = \frac{\hbar}{c}$:

$$\phi(0,0,\hbar/c) = \frac{1}{(2\pi\hbar)^{3/2}} \int dx\, dy\, dz\; \psi(x,y,z)\, e^{-iz/c}$$

Insert $\psi(x,y,z) = N e^{-\left(\frac{|x|}{2a} + \frac{|y|}{2b} + \frac{|z|}{2c}\right)}$:

$$\phi = \frac{N}{(2\pi\hbar)^{3/2}} \left(\int dx\, e^{-\frac{|x|}{2a}}\right) \left(\int dy\, e^{-\frac{|y|}{2b}}\right) \left(\int dz\, e^{-\frac{|z|}{2c}} e^{-iz/c}\right)$$

Using

$$\int_{-\infty}^{\infty} e^{-\frac{|x|}{2a}}\, dx = 4a, \quad \int_{-\infty}^{\infty} e^{-\frac{|y|}{2b}}\, dy = 4b$$

and splitting the z-integral:

$$\int_{-\infty}^{\infty} dz\, e^{-\frac{|z|}{2c}} e^{-iz/c} = \int_{-\infty}^{0} dz\, e^{\frac{z}{2c}} e^{-iz/c} + \int_{0}^{\infty} dz\, e^{-\frac{z}{2c}} e^{-iz/c}$$

$$= \int_{-\infty}^{0} dz\, e^{\left(\frac{1}{2c} - \frac{i}{c}\right)z} + \int_{0}^{\infty} dz\, e^{-\left(\frac{1}{2c} + \frac{i}{c}\right)z}$$

Evaluating:

$$= \frac{2c}{1-2i} + \frac{2c}{1+2i} = \frac{4c}{1+4} = \frac{4c}{5}$$

Therefore:

$$\phi(0,0,\hbar/c) = \frac{N}{(2\pi\hbar)^{3/2}} \cdot (4a)(4b) \cdot \frac{4c}{5} = \frac{64abc\, N}{5(2\pi\hbar)^{3/2}}$$

Using $N = \frac{1}{\sqrt{8abc}}$:

$$\phi = \frac{64abc}{5(2\pi\hbar)^{3/2}} \cdot \frac{1}{\sqrt{8abc}} = \frac{64}{5(2\pi\hbar)^{3/2}} \sqrt{\frac{abc}{8}}$$

Probability:

$$P = |\phi|^2\, dp_x dp_y dp_z = \frac{64}{25} \frac{abc}{(\pi\hbar)^3}\, dp_x dp_y dp_z$$

11. Operator Algebra and Measurements Consider a three-dimensional Hilbert space with an orthonormal basis $\{\phi_1,\phi_2,\phi_3\}$. The representations of two operators in this basis are

$$H = \hbar\omega_0 \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad B = b \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

(a) Compute the commutator of H and B.

Calculation

We compute

$$[H,B] = HB - BH.$$

Since

$$H = \hbar\omega_0 \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad B = b \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

we obtain

$$HB = \hbar\omega_0 b \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix},$$

$$BH = \hbar\omega_0 b \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Hence

$$[H,B] = 0.$$

(b) Give the common eigenbasis of H and B.

Calculation

The operator H is already diagonal in the basis $\{\phi_1,\phi_2,\phi_3\}$:

$$H\phi_1 = \hbar\omega_0\phi_1, \qquad H\phi_2 = -\hbar\omega_0\phi_2, \qquad H\phi_3 = -\hbar\omega_0\phi_3.$$

For B, we have

$$B\phi_1 = b\phi_1,$$

and on the subspace spanned by ϕ_2,ϕ_3,

$$B_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its eigenvalues are $\lambda = \pm 1$, with normalized eigenvectors

$$v_2 = \frac{1}{\sqrt{2}}(\phi_2 + \phi_3), \qquad v_3 = \frac{1}{\sqrt{2}}(\phi_2 - \phi_3).$$

Also

$$v_1 = \phi_1.$$

Therefore a common eigenbasis of H and B is

$$\begin{aligned} v_1 &= \phi_1, \\ v_2 &= \frac{1}{\sqrt{2}}(\phi_2 + \phi_3), \\ v_3 &= \frac{1}{\sqrt{2}}(\phi_2 - \phi_3). \end{aligned}$$

(c) Do the pairs $\{H,B\}$ and $\{H^2,B\}$ form a complete set of commuting observables?

Calculation

Since $[H,B] = 0$ and both operators are Hermitian, they are compatible observables. For the pair $\{H,B\}$:

$$\begin{aligned} Hv_1 &= \hbar\omega_0 v_1, & Bv_1 &= bv_1, \\ Hv_2 &= -\hbar\omega_0 v_2, & Bv_2 &= bv_2, \\ Hv_3 &= -\hbar\omega_0 v_3, & Bv_3 &= -bv_3. \end{aligned}$$

The simultaneous eigenvalue pairs are therefore

$$(\hbar\omega_0,b), \qquad (-\hbar\omega_0,b), \qquad (-\hbar\omega_0, -b),$$

which distinguish all basis vectors. Hence $\{H,B\}$ is a complete set of commuting observables.
For the pair $\{H^2,B\}$:

$$H^2 = \hbar^2\omega_0^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \hbar^2\omega_0^2 I.$$

Thus H^2 has the same eigenvalue for all states and gives no additional information.

Therefore $\{H^2,B\}$ is not complete.

(d) The system is in the state

$$\psi = \left\{\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2}\right\}.$$

Compute the probabilities that:

i. a measurement of H gives the result $\hbar\omega_0$,

ii. a measurement of H gives the result $-\hbar\omega_0$,

iii. a measurement of B gives the result b,

iv. a measurement of B gives the result $-b$,

v. a measurement of B gives the result $3b$.

Calculation

In the basis $\{\phi_1,\phi_2,\phi_3\}$,

$$\psi = \frac{1}{\sqrt{2}}\phi_1 + \frac{1}{2}\phi_2 + \frac{1}{2}\phi_3.$$

For H:

$$\begin{aligned} P(H = \hbar\omega_0) &= \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}, \\ P(H = -\hbar\omega_0) &= \left|\frac{1}{2}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{1}{2}. \end{aligned}$$

For B, rewrite ψ in the common eigenbasis:

$$\begin{aligned} \phi_2 &= \frac{1}{\sqrt{2}}(v_2 + v_3), \\ \phi_3 &= \frac{1}{\sqrt{2}}(v_2 - v_3). \end{aligned}$$

Therefore

$$\begin{aligned} \psi &= \frac{1}{\sqrt{2}}v_1 + \frac{1}{2}\cdot\frac{1}{\sqrt{2}}(v_2 + v_3) + \frac{1}{2}\cdot\frac{1}{\sqrt{2}}(v_2 - v_3) \\ &= \frac{1}{\sqrt{2}}v_1 + \frac{1}{\sqrt{2}}v_2. \end{aligned}$$

So

$$\begin{aligned} P(B = b) &= \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 = 1, \\ P(B = -b) &= 0, \\ P(B = 3b) &= 0, \end{aligned}$$

since $3b$ is not an eigenvalue of B.

(e) The system from (d) is measured twice.

i. In the first measurement of H, one obtains $-\hbar\omega_0$. Compute the probability that a second measurement of B gives $+b$.

ii. In the first measurement of H, one obtains $-\hbar\omega_0$. Compute the probability that a second measurement of B gives $-b$.

iii. In the first measurement of H, one obtains $\hbar\omega_0$. What is obtained in the second measurement of B?

Calculation

If the first measurement of H yields $-\hbar\omega_0$, the state collapses onto the normalized projection of ψ onto the eigenspace spanned by ϕ_2,ϕ_3:

$$\psi' = \frac{\frac{1}{2}\phi_2 + \frac{1}{2}\phi_3}{\sqrt{\frac{1}{4}+\frac{1}{4}}} = \frac{1}{\sqrt{2}}(\phi_2 + \phi_3) = v_2.$$

Since v_2 is an eigenstate of B with eigenvalue $+b$:

$$P(B = +b) = 1, \qquad P(B = -b) = 0.$$

If the first measurement of H yields $\hbar\omega_0$, the state collapses to

$$\psi'' = \phi_1 = v_1.$$

Since v_1 is an eigenstate of B with eigenvalue $+b$, the second measurement yields

$$B = +b$$

with certainty.

12. States of Minimal Position and Momentum Uncertainty The Heisenberg uncertainty relation for position and momentum operators can be proven as follows. For a given state $|\psi\rangle$ define

$$\hat{X}' = \hat{X} - \langle\hat{X}\rangle, \qquad \hat{P}' = \hat{P} - \langle\hat{P}\rangle,$$

such that

$$\Delta X^2 = \langle\hat{X}'^2\rangle, \qquad \Delta P^2 = \langle\hat{P}'^2\rangle.$$

Consider the state

$$|\phi\rangle = (\hat{X}' + i\lambda\hat{P}')\,|\psi\rangle$$

with arbitrary real parameter λ.

(a) Show that from

$$\forall \lambda \in \mathbb{R} : \langle \phi | \phi \rangle \geq 0$$

the Heisenberg uncertainty relation $\Delta X \Delta P \geq \hbar/2$ follows.

Calculation

We have

$$\langle \phi | \phi \rangle = \langle \psi | \, (\hat{X}' - i\lambda \hat{P}')(\hat{X}' + i\lambda \hat{P}') \, | \psi \rangle \geq 0.$$

Expanding:

$$\langle \phi | \phi \rangle = \langle \psi | \, \hat{X}'^2 \, | \psi \rangle + i\lambda \, \langle \psi | \, \hat{X}' \hat{P}' \, | \psi \rangle - i\lambda \, \langle \psi | \, \hat{P}' \hat{X}' \, | \psi \rangle + \lambda^2 \, \langle \psi | \, \hat{P}'^2 \, | \psi \rangle \, .$$

Hence

$$\langle \phi | \phi \rangle = \Delta X^2 + \lambda^2 \Delta P^2 + i\lambda \langle [\hat{X}', \hat{P}'] \rangle.$$

Since constants commute,

$$[\hat{X}', \hat{P}'] = [\hat{X}, \hat{P}] = i\hbar,$$

therefore

$$\begin{aligned} \langle \phi | \phi \rangle &= \Delta X^2 + \lambda^2 \Delta P^2 + i\lambda (i\hbar) \\ &= \Delta X^2 + \lambda^2 \Delta P^2 - \lambda \hbar. \end{aligned}$$

Thus for all $\lambda \in \mathbb{R}$,

$$\Delta X^2 + \lambda^2 \Delta P^2 - \lambda \hbar \geq 0.$$

This quadratic expression in λ must be non-negative for all real λ, so its discriminant must satisfy

$$\hbar^2 - 4 \Delta X^2 \Delta P^2 \leq 0.$$

Hence

$$4 \Delta X^2 \Delta P^2 \geq \hbar^2,$$

and therefore

$$\Delta X \Delta P \geq \frac{\hbar}{2}.$$

(b) Show that equality holds for a state $|\psi\rangle$ if and only if

$$(\hat{X}' + i\lambda_0 \hat{P}')\,|\psi\rangle = 0$$

with

$$\lambda_0 = \frac{\hbar}{2\Delta P^2} = \frac{2\Delta X^2}{\hbar}.$$

Calculation

Equality in the uncertainty relation holds if and only if the quadratic expression

$$f(\lambda) = \Delta X^2 + \lambda^2 \Delta P^2 - \lambda\hbar$$

has a double root, i.e. its minimum is zero.
The minimum is attained at

$$\frac{df}{d\lambda} = 2\lambda\Delta P^2 - \hbar = 0 \qquad \Longrightarrow \qquad \lambda_0 = \frac{\hbar}{2\Delta P^2}.$$

At equality we also have

$$\Delta X \Delta P = \frac{\hbar}{2},$$

so that

$$\Delta X^2 = \frac{\hbar^2}{4\Delta P^2}.$$

Hence

$$\lambda_0 = \frac{\hbar}{2\Delta P^2} = \frac{2\Delta X^2}{\hbar}.$$

Now, if

$$(\hat{X}' + i\lambda_0 \hat{P}')\,|\psi\rangle = 0,$$

then $|\phi\rangle = 0$ and therefore

$$\langle\phi|\phi\rangle = 0,$$

so the lower bound is saturated.
Conversely, if equality holds, then the minimum value of $\langle\phi|\phi\rangle$ is zero at $\lambda = \lambda_0$, which implies

$$\|(\hat{X}' + i\lambda_0 \hat{P}')\,|\psi\rangle\,\|^2 = 0.$$

Therefore

$$(\hat{X}' + i\lambda_0 \hat{P}')\,|\psi\rangle = 0.$$

(c) Write the equation

$$(\hat{X}' + i\lambda_0 \hat{P}')\,|\psi\rangle = 0$$

in the position representation and solve the resulting differential equation for $\psi(x) = \langle x|\psi\rangle$. Consider the ansatz

$$\psi(x) = \exp\left(i\frac{\langle \hat{P}\rangle x}{\hbar}\right) f(x - \langle \hat{X}\rangle).$$

Which states are obtained?

Calculation

In the position representation,

$$\hat{X}\psi(x) = x\psi(x), \qquad \hat{P}\psi(x) = -i\hbar\frac{d}{dx}\psi(x).$$

Therefore

$$\hat{X}' = x - \langle \hat{X}\rangle, \qquad \hat{P}' = -i\hbar\frac{d}{dx} - \langle \hat{P}\rangle.$$

The equation becomes

$$\left[x - \langle \hat{X}\rangle + i\lambda_0\left(-i\hbar\frac{d}{dx} - \langle \hat{P}\rangle\right)\right]\psi(x) = 0.$$

Hence

$$\left[x - \langle \hat{X}\rangle + \lambda_0\hbar\frac{d}{dx} - i\lambda_0\langle \hat{P}\rangle\right]\psi(x) = 0.$$

Use the ansatz

$$\psi(x) = \exp\left(i\frac{\langle \hat{P}\rangle x}{\hbar}\right) f(x - \langle \hat{X}\rangle).$$

Let

$$\xi = x - \langle \hat{X}\rangle.$$

Then

$$\frac{d\psi}{dx} = \exp\left(i\frac{\langle\hat{P}\rangle x}{\hbar}\right)\left(i\frac{\langle\hat{P}\rangle}{\hbar}f(\xi) + f'(\xi)\right).$$

Substituting into the differential equation, the terms containing $\langle\hat{P}\rangle$ cancel, and we obtain

$$\xi f(\xi) + \lambda_0\hbar f'(\xi) = 0.$$

Thus

$$\frac{f'(\xi)}{f(\xi)} = -\frac{\xi}{\lambda_0\hbar}.$$

Integrating:

$$\ln f(\xi) = -\frac{\xi^2}{2\lambda_0\hbar} + \text{const}.$$

Therefore

$$f(\xi) = C\exp\left(-\frac{\xi^2}{2\lambda_0\hbar}\right).$$

Since

$$\lambda_0 = \frac{2\Delta X^2}{\hbar},$$

we get

$$f(\xi) = C\exp\left(-\frac{\xi^2}{4\Delta X^2}\right).$$

Hence

$$\psi(x) = C\exp\left(-\frac{(x-\langle\hat{X}\rangle)^2}{4\Delta X^2} + i\frac{\langle\hat{P}\rangle x}{\hbar}\right).$$

To normalize:

$$\begin{aligned} 1 &= |C|^2\int_{-\infty}^{\infty}\exp\left(-\frac{(x-\langle\hat{X}\rangle)^2}{2\Delta X^2}\right)dx \\ &= |C|^2\sqrt{2\pi}\,\Delta X. \end{aligned}$$

Thus

$$C = \frac{1}{(2\pi\Delta X^2)^{1/4}}$$

up to an irrelevant overall phase.

Therefore the states are

$$\psi(x) = \frac{1}{(2\pi\Delta X^2)^{1/4}} \exp\left(-\frac{(x - \langle \hat{X} \rangle)^2}{4\Delta X^2} + i\frac{\langle \hat{P} \rangle x}{\hbar} \right),$$

i.e. Gaussian wave packets, also called minimum-uncertainty states.